KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN MECHANICAL ENGINEERING

EMM 312: FLUID MECHANICS III

1^{ST} SEMESTER 2022/2023 TIME: 2 Hours

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INSTRUCTIONS:

1. **This paper contains FIVE questions.**
2. **Question ONE is compulsory and carries 30 marks.**
3. **FOUR remaining questions carry 20 marks each.**
4. **Attempt any TWO of the remaining Four questions**

QUESTION ONE (Compulsory) - 30 marks

a) i. Determine the ratio of depth of the flow Dm to the breadth Bm in a rectangular open channel which will make discharge maximum for a given cross section area (excavation), slope and roughness coefficient.
Assume steady uniform flow and that Chezy circuit C is constant.

ii. A channel of rectangular cross-section conveys water at the rate of $9.2m^3$/sec flowing at a velocity of 1.25m/sec.
Find the gradient required for uniform steady flow if the cross section area is designed for maximum discharge.

b) i. Drive an expression for the critical velocity through any cross-sectional channel in terms of discharge (Q), cross sectional area (A) and width of the water surface (B).

ii. Hence show that in a rectangular channel, the critical depth is two thirds of the Specific Energy Head.

c) Water flows in a channel of a rectangular cross-section at a velocity of 2.2m/sec and a depth of 1.8m.
Calculate the

i. Specific energy of the flow
ii. Critical depth
iii. Discharge under critical flow conditions if the channel is 3.6m wide.

QUESTION TWO (20 marks)

a) Air from a large vessel discharges into the atmosphere through a small orifice in its side. Working from first principles, stating the assumptions made and explaining the meaning of all symbols used, show that the mass flow rate through the orifice $\dot{m}$ is

Given by $m' = Cde_1 a\sqrt{2\left(\frac{\gamma}{\gamma-1}\right)\frac{p_1}{e}(1-r_p^{\gamma-1})r_{p\gamma}^{2}}$ $\quad V_p \therefore \frac{2}{\gamma}$

Where Cd = coefficient of discharge for the orifice

e_1 =mass density of air in the vessel

a=cross-sectional area of the orifice.

r=expression/compression adiabatic index

P_1=pressure in the vessel

γ_p=compression ratio

$= \frac{p_2}{p_1}$

b) The pressure and temperature of the air in the vessel are 207KN/m^2 absolute and 15^{0}C respectively. The actual mass flow rate through the orifice is 0.154kg/sec. Calculate the diameter of the orifice in mm.

Assume

$\gamma = 1.4$

$R = 287\, J/kg$ ok

Atmospheric pressure corresponds to 775.77mm of mercury

Cd = 0.64

QUESTION THREE (20 MARKS)

(a) Show that the mass flow rate ($\dot{m}$) through a horizontal venturimeter of a compressible fluid under adiabatic conditions is given by:

$$\dot{m} = C_d a_1 \frac{p_1}{RT_1}\sqrt{\frac{2\left(\frac{\gamma}{\gamma-1}\right)RT_1\left[1-r_p^{\frac{\gamma-1}{\gamma}}\right]}{\left(\frac{a_1}{a_2}\right)^2\left(\frac{1}{rp}\right)^{2/\gamma}+1}}$$

When Cd = coefficient of discharge

a_1 = cross-sectional area of the pipe

p_1 = absolute pressure in the pipeline

R = universal gas constrant

T_1=absolute temperature of the fluid in the pipe

a_2 = cross sectional area of the throat

r_p = compression ratio

γ = adiabatic index of compression/expansion

(b) A horizontal venturi meter 400 mm in diameter at inlet and 200 mm at the throat is used for measuring the flow of air. The pressure at the inlet and the throat are $150 KN/m^2$ and $125 KN/m^2$ absolute respectively while the temperature at inlet is 18^0 c.

Calculate the discharge in kg/sec

Assume: $\gamma = 1.4$

$R = 287 \, J/kg^0 K$

$C_d = 0.97$

QUESTION FOUR (20 MARKS)

(a) Concisely discuss the occurrence of the water hammer phenomenon in pipelines.

(b) During a water hammer phenomenon in a pipeline resulting from a sudden complete closure of a valve, show that the pressure rise is given by

$$P = V\sqrt{\frac{e}{\frac{1}{k} + \frac{d(5-4v)}{4tE}}}$$

Where v = velocity of flow before closure of the value

d = internal diameter of the pipe

t = thickness of the pipe

E = modulus of Elasticity of the pipe material

k = bulk modulus of water

V = Poisson's ratio

e = mass density of water in the pipeline.

(c) Water flows through a horizontal pipeline at a maximum discharge of $0.2097 \, m^3/sec$. While the maximum pressure rise is $1500 \, kN/m^2$. Determine the internal diameter of the pipe.

Assume: The pipe is rigid

For water $k = 207 \times 10^7 N/m^2$

QUESTION FIVE (20 MARKS)

(a) Distinguish between Normal and Oblique shock waves

(b) Show that for a normal shock wave.

$$p_1\left(1+\gamma M_{a_1}^2\right)=p_2\left(1+\gamma M_{a_2}^2\right)$$

Where p_1 and p_2 are pressures upstream and downstream the shock wave Ma_1 and Ma_2 are the Mach Numbers upstream and downstream of the shock waves respectively.

γ = adiabatic index

(c) A projectile with rounded nose moves through still air at M_a=5. The air pressure and temperature are 60kN/m^2 and -10^0 c respectively. Assuming that the Shock wave at the nozzle of the projectile is detached and normal to it,

Determine the stagnation

(i) Pressure and
(ii) Temperature at the nose
Assume γ =1.4